

Fusion Science and Technology

ISSN: 1536-1055 (Print) 1943-7641 (Online) Journal homepage: www.tandfonline.com/journals/ufst20

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To cite this article: Wei Zhao, Yali Wang, Yuzhong Jin, Li Zhao, Hongxia Zhou, Lin Nie, Guangwu Zhong, Chunjia Liu, Christopher Watts & James Paul Gunn (2020) Preliminary Design of ITER Divertor Langmuir Probe System, Fusion Science and Technology, 76:2, 79-87, DOI: 10.1080/15361055.2019.1674123

To link to this article: https://doi.org/10.1080/15361055.2019.1674123

Published online: 22 Nov 2019.

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FUSION SCIENCE AND TECHNOLOGY · VOLUME 76 · 79–87 · FEBRUARY 2020
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DOI: https://doi.org/10.1080/15361055.2019.1674123

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Preliminary Design of ITER Divertor Langmuir Probe System

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Received March 7, 2019
Accepted for Publication September 16, 2019

Abstract — *The primary aim of the ITER divertor Langmuir probe system is to measure the plasma parameters at the divertor target plates. Saturation ion flux coming from the direct-current biased probe mode is used for advanced machine control, and the swept double-probe mode is recommended to measure electron temperature and density for physics studies. The design of the probe system includes three parts. First, tungsten Langmuir probes are mounted on the side of the target plates for collecting current from plasma, and thermomechanical simulation results show the design of the probe is robust and can survive under harsh working environments. Second, the electronics consists of the power supply, mode switching, and signal conditioning box and is used for driving Langmuir probes in different operation modes to obtain expected plasma information. Third, the functions of instrument and control include publishing configuration; monitoring and control; calibration; data acquisition; communication with the control, data access and communication (CODAC) system; and real-time ion flux measurement at the divertor target. The system design also complies with ITER's technical practices, standards, and codes.*

Keywords — *ITER, divertor, Langmuir probe, design.*

Note — *Some figures may be in color only in the electronic version.*

I. INTRODUCTION

Currently under construction, ITER will be the biggest tokamak the world over for scientific and engineering research and perhaps the most challenging scientific endeavor being undertaken today.[1] There are tens of diagnostic devices in the ITER tokamak,[2] and divertor Langmuir probes (DLPs) are used to measure the plasma parameters at the divertor target plates. This measurement is used for machine control and physics study. For machine control, it is envisioned that the probes will provide a signal indicating the attached/detached state of the divertor plasma, while for physics study, the system will need to supply more detailed measurements of the electron density and temperature. The Langmuir probe system consists of DLPs and back-end electronics, which includes power supply, probe operation mode switching box, signal conditioning unit, and instrument and control (I&C). The back-end electronics is located in level B1 of the Diagnostics Building. The main functions of the electronics are Langmuir probe driving, operation mode switching, and signal transmission and conditioning, and the four-quadrant power output is mandated to drive the Langmuir probes. The I&C of the Langmuir probe system complies with the requirements and can work perfectly on communication, data acquisition, configuration publishing, and various monitoring and control functions. DLP brazed on the monoblock includes a tungsten probe, insulation ceramic, and tungsten shield, and all DLP components are

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TABLE I

Measurement Requirements from Memorandums and Annex B

Measurement	Parameter	Operational Role	Latency (Real Time)	Parameter Range	Specified Time Resolution	Specified Spatial Resolution	Specified Accuracy
Plasma parameter at the divertor targets	Γ_i, divertor ion flux	Advanced control	2.5 ms	10^5 to 10^7 A/m^2	10 ms	12 mm	10% relative
Plasma parameter at the divertor targets	n_e, divertor target	Physics	—	10^{18} to 10^{22} m^{-3}	1 ms	12 mm	30%
	T_e, divertor target		—	1 to 150 eV			30%
	Γ_i, divertor ion flux		—	10^5 to 10^7 A/m^2	10 μs		30%

welded connections. From the probes to the back-end electronics, many cables are employed to drive the probes and transmit probe signals.

II. ITER LANGMUIR PROBE SYSTEM MEASUREMENT REQUIREMENTS

According to the ITER project requirements and diagnostics system requirements document (unpublished), the primary contribution of the Langmuir probe system is to provide the electron temperature and density for advanced control, and the supplementary contribution to basic control is to measure the plasma position and shape at the divertor targets. More careful and detailed work from Annex B and the memorandums "Measurement Requirements and Flow-Down for the Divertor Plasma Parameters Diagnostics" and "Probe Measurement Estimates for Tungsten Divertor" (unpublished) elaborates on these measurement requirements for both advanced control and physics study, and at the same time, the documents also give the hardware requirements, such as the probe driving voltage and current from the plasma. Based on these documents, the comprehensive measurement requirements of the ITER Langmuir probe system are defined and summarized in Table I.

III. DIVERTOR LANGMUIR PROBE DESIGN

Langmuir probes, or electrostatic probes, are the first plasma diagnostic and were developed by Langmuir in the 1920s. The principle is relatively simple: A conductor in contact with the plasma is biased with a voltage V relative to the local plasma potential. The current I drawn by the probe is proportional to the local plasma density and temperature and in the simplest approximation is

$$I \propto I_e e^{V/T_e} - I_i \,, \tag{1}$$

where

I_e = electron current

I_i = ion current

T_e = electron temperature

V = potential on probe.

The actual relationship among these parameters is quite a bit more complex, depending on the detailed probe geometry, presence of magnetic field, potential, etc.

Langmuir probes have been successfully implemented in nearly every tokamak in the world, such as HL-2A (Ref. 3), EAST (Ref. 4), Tore Supra (Ref. 5), DIII-D (Ref. 6), JET (Ref. 7), and JT60-SA (Ref. 8). In ITER, the primary aim of the Langmuir probes is to measure the plasma parameters at the divertor target plates, including electron temperature, electron density, and ion flux, and these parameters are used for advanced machine control and physics study. For machine control, a simple and robust control signal ion saturation current (jsat) is the clear preferred choice. During normal operation, jsat will provide a clear, prompt indication of plasma attachment, which can be used as a basis for machine control, while for physics study, the standard swept single-ended Langmuir probe configuration and swept double probes are used to measure the electron temperature and density. As single-ended Langmuir operation is the standard configuration for most Langmuir probes, the signal provides an indication of the local plasma parameters, with the caveat that the interpretation of the signal characteristic is not straightforward and is often controversial. Moreover, when sweeping toward the electron saturation end of the characteristic, there is a danger of drawing too much current, possibly damaging probes. Swept double-probe mode operation refers to two floating probes relative to each other. The probe characteristic is more easily interpreted than that of a swept single-ended probe, and both plasma

density and temperature can be extracted from the data. As with jsat operation, this configuration also ensures that the probes do not draw excessive current. Therefore, the direct-current (DC) biased probe mode and swept double-probe mode are recommended for the operation of ITER Langmuir probes.

ITER Organization has chosen to operate using an all-tungsten divertor.[9] This scenario requires a full tungsten divertor that will need to survive the first helium and hydrogen plasma campaigns and be well into the initial nuclear deuterium-tritium phase. So, the probe design work has been carried out assuming a full tungsten divertor, and flush-mounted probes are used in the ITER tokamak.[10,11] Figure 1 depicts a sketch of the array on one divertor cassette. Four hundred probes are attached on 5 of the 54 divertor cassettes over three toroidal positions separated by 120 deg in Fig. 2.

The probes are subjected to high, steady-state heat loads from plasma flux, photon irradiation, and energetic neutrals from charge exchange reactions, which are a new issue with respect to existing tokamaks. First, the plasma load that the SOLPS simulations indicate can reach up to 15 MW/m^2 at the strike point of the divertor.[12] However, we have chosen a more conservative consideration in design; the maximum divertor engineering value of 20 MW/m^2 for up to 10-s exposure has typically been assumed.[13] Second, the probe is subjected to intense electromagnetic radiation, up to 4 MW/m^2 (Ref. 14). Finally, there will be volumetric heating and radiation damage from fusion neutrons. Therefore, a robust design to survive under such severe conditions is required.

The following picture is the present design of the ITER divertor Langmuir probe, which consists of a 2-mm-diameter tungsten probe, a complex shape tungsten shield, and an alumina coating between the tungsten probe and shield. All the components are bonded together to make a high thermal transferring capacity in order to reduce the

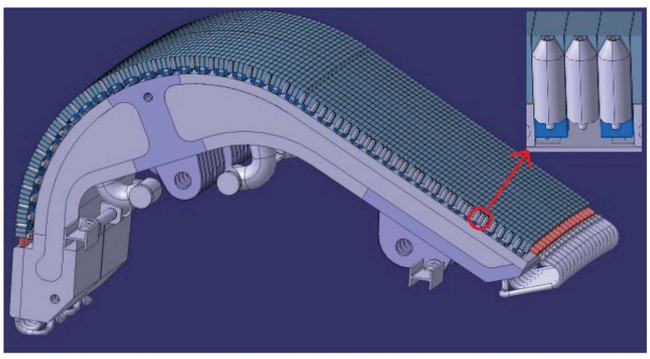


Fig. 1. Langmuir probe along divertor target.

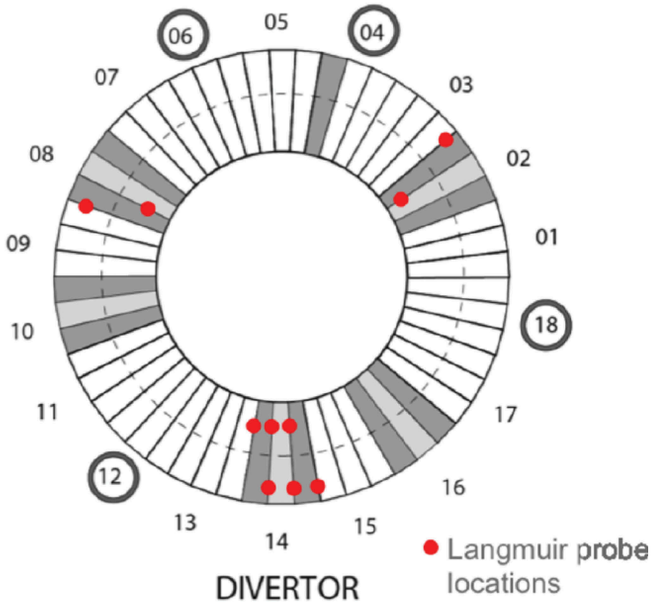


Fig. 2. Langmuir probe distribution in ITER.

temperature of the probe. The relevant manufacturing technology, machining, sintering, and brazing are being developed at Xia Men Tungsten Company, Ltd., China (XTC). Combining the above heat loads, cooling condition, and emission, the complete boundary conditions are determined for thermal simulation. In addition, the properties of the materials used for the thermal analysis are from the ITER material handbook, and the measurements are from XTC. The thermal analysis results of the Langmuir probe under steady-state working conditions are shown in Fig. 3. The maximum temperature of the monoblock is about 1533°C, which is located at the top surface close to the DLP out of the monoblock chamfer. The maximum temperatures of the probe body and alumina are about 1173°C and 643°C, respectively. A simple summary of all the parts at maximum temperature under steady-state and slow transient-state working conditions is given in Table II.

A corresponding mechanical analysis is also carried out. The stress of the tungsten probe and tungsten shield is lower than the strength of the W material, as shown in Figs. 4a and 4b. The stress of the ceramic in a very small zone exceeds the alumina allowable strength. Figure 4c shows the details of the stress distribution of the alumina tube, as the optimization work of the DLP design is ongoing, and this problem will be solved in the future.

IV. ELECTRONICS DESIGN OF LANGMUIR PROBE SYSTEM

As previously introduced, the DLP system consists of Langmuir probes in the tokamak and back-end electronics

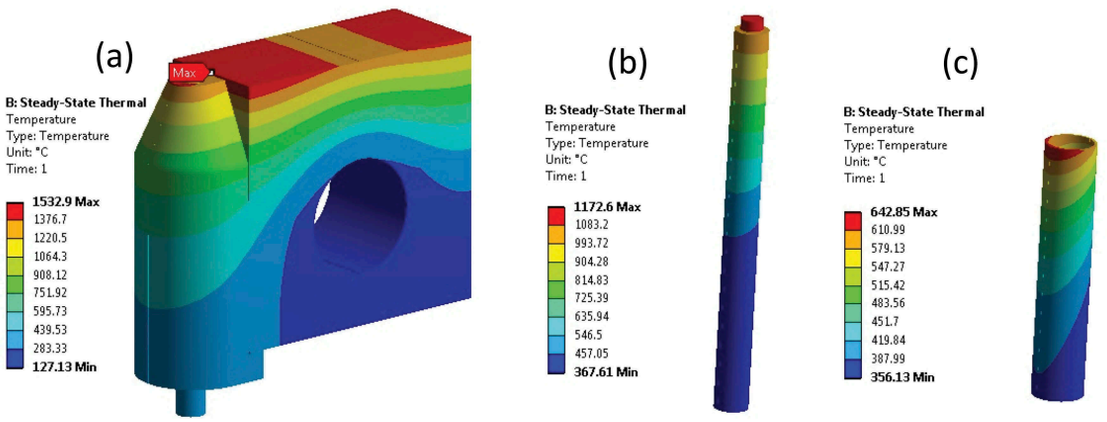


Fig. 3. (a) Temperature field of DLP and monoblock, (b) temperature field of probe, and (c) temperature field of ceramic.

TABLE II

Temperature of Each Part Under Different Working Conditions

Heat Flux/Maximum Temperature (°C)	10 MW/m^2	20 MW/m^2
Tungsten probe	1173	1815
Alumina	643	849
Tungsten shield	1404	1649
Monoblock	1533	2498

housed in 24 cubicles at the diagnostic area. In the cubicle layout, shown in Fig. 5, there is only one type A cubicle; the others are type B. The cubicle contains a standard ITER network switch, a plant system host (PSH) industrial computer (only in the type A cubicle), fast controllers [high-performance computer (HPC), PXIe chassis, and PXIe modules], power supplies, a mode switching and signal conditioning box (MSSCB), and a slow controller.[15] To categorize the DLP system simply, we call the power supply and MSSCB "electronics"; we call the other parts "I&C." So, the back-end electronics can be simply separated into electronics and I&C. The fast controller can communicate with the power supply and MSSCB via RS-485 and can send analog signals to the power supply and obtain voltage and current signals from the MSSCB used for probe operation mode switching and signal conditioning.

According to the Langmuir probe principle, plasma is not always a resistor; sometimes it is a source. This means that the current collected and the voltage applied may not always be in the same phase, so a four-quadrant power supply is mandated to drive the probes. In ITER, flush-mounted probes are employed to collect the ion current. The effective collecting area of the probe is the sum of its geometrical projection along the magnetic field lines plus the projection of the sheath that expands above it when biased negatively with respect to the surrounding target. The effective collecting area can be written as[16,17]

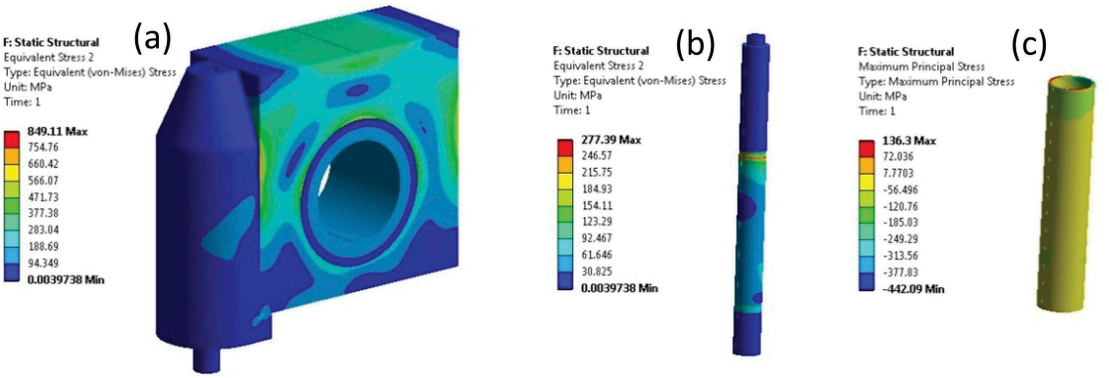


Fig. 4. (a) Stress field of DLP and monoblock, (b) stress field of probe, and (c) stress field of ceramic.

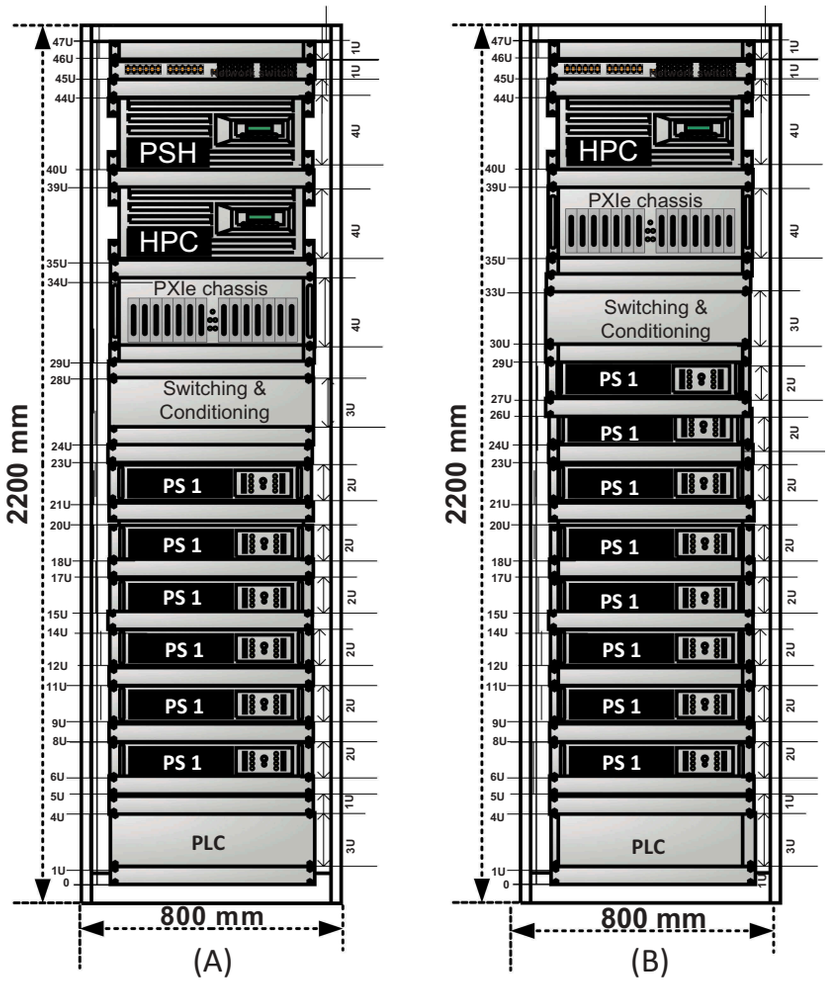


Fig. 5. Cubicle layout of DLP system.

$$A_{eff} = A_{probe}\sin\alpha + W_{probe}\frac{0.5}{\sqrt{\sin\alpha}}\lambda_D\left[\left(\frac{V_{plasma}-V_{bias}}{T_e}\right)^{3/4} - \left(\frac{V_{plasma}}{T_e}\right)^{3/4}\right] \quad (2)$$

and

$$I = n_e e c_s A_{eff}\ , \quad (3)$$

where

A = area

α = angle between probe and magnetic field

W = probe width

λ_D = Debye length

V_{plasma} = plasma potential

V_{bias} = probe potential to ground

T_e = electron temperature

c_s = ion sound speed

I = ion current of probe.

Referring to the divertor design, the expected plasma parameters of the electron densities can be up to a few 10^{21} m^{-3} and electron temperatures up to about 40 eV at the target plates. The calculations were made using the SOLPS-EIRENE code package, provided by Kukushkin et al.[12] Finally, we can calculate the total ion current collected for a large negative bias. The maximum expected current is less than 1 A/probe.

As the commercial power supply cannot meet the DLP system requirements and ITER special design specifications, a new power supply is designed that has these good basic properties:

1. The current and voltage of the power supply output are 2.5 A and 175 V, respectively.

2. The power supply can be remotely controlled and communicate with the fast controller.

3. The power supply bandwidth can be from DC to 15 kHz, ensuring good DC bias, sweeping wave, and quick responding.

4. The power supply is small with high efficiency, and the height is less than 2 U.

Figure 6 is a simplified schematic diagram of the power supply. From left to right, the modules are alternating-current (AC)–DC (AC-DC) module, energy absorption and consumption circuit, H bridge circuit, and filter, and they function as follows:

1. *AC-DC module*: This module gets electricity from the distribution panel, and then AC power is transmitted to DC power, which provides power to the H bridge circuit.

2. *Energy absorption and consumption circuit*: Since plasma sometimes behaves like a current source, the absorption module is used to consume energy from the plasma.

3. *H bridge circuit and filter*: The switch mode amplifier using pulse width modulation technology produces a variable duty cycle pulse train that acts as input; the power stage taking the form of the pulse train combined with filter outputs the expected waveform.

Since one power supply drives two probes, in order to satisfy the recommended DC biased and the swept double-mode operation of the Langmuir probes, the power supply current output is defined as 2.5 A, and the output voltage is ±175 V with arbitrary offset. As is well known, the plasma behavior is very complex with certain destructive events, such as edge-localized modes (ELMs), which brings high temperature and density plasma to hit the divertor and probes, while the current can abruptly reach several amperes, so the power supply is designed to withstand 10 A in a few milliseconds in order to allow the Langmuir probes to measure the ELMy without cutting off. But, if the current is very large, the protection mechanism of the power supply will be triggered.

As the ITER Langmuir probes are installed on divertor targets in the vacuum vessel, they are far from the diagnostic area, which is the location of the back-end electronics. In order to interpret the probe signal, the measured probe potential must refer to the nominal plasma ground. It is not easy to do, especially in plasmas with significant noise. Commonly, the reference of the probes may be the local tokamak metal surface near the probes or the building ground of the power supply. In order to add flexibility, several grounds are proposed at the divertor cassette: Three grounds are on the inner, outer, and center of the cassette, and another one is close to the cassette attachment point to the vacuum vessel. These grounds are to be viewed as signals and will not be tied to the local ground at the location of the electronics. Which ground to be used is determined by experiment results.

The ITER Langmuir probe system, like other diagnostic systems, is to be a plant system to the ITER control, data access and communication (CODAC) system to allow ITER operators to remotely access and control it. The system includes several functions: communication, data acquisition, configuration publishing, and various monitoring and control functions.

As a CODAC I&C requirement, all probe electronics will be connected to CODAC via a middle layer that includes the PSH, slow controller, and fast controller.

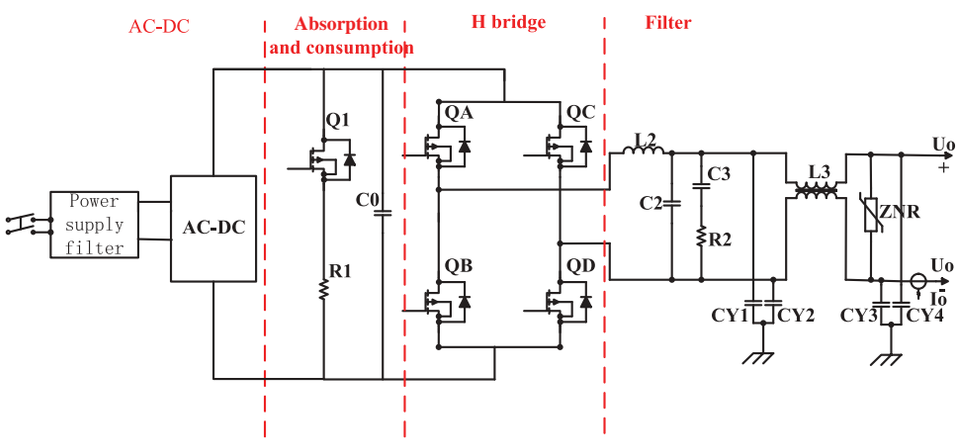


Fig. 6. Topology diagram of power supply.

The detailed architecture corresponding to the cubicle layout and network configuration are shown in Figs. 7 and 8 (Ref. 18). The PSH, which is connected to the plant operation network (PON) and time communication network (TCN), provides asynchronous communication from the CODAC system to the DLP plant control system and vice versa and also provides command dispatching, state monitoring, data flow, and configuration functions. The fast controller and slow controller are local units in charge of implementing the functional and physical part of the control and data acquisition of the plant system. The slow controller only connected to PON is responsible for implementation of multiple functions like reading slow signals from the sensors and monitoring the cubicle health via the Siemens S7-series PLC, which is considered the standard solution for slow controllers in ITER. The fast controller is connected to the PON, TCN, synchronous databus network (SDN), and data archiving network (DAN). The main functions of fast controller are alarm handling, event handling, logging, data communication functions, data acquisition, data analysis, signal monitoring, and controller of the power supply.

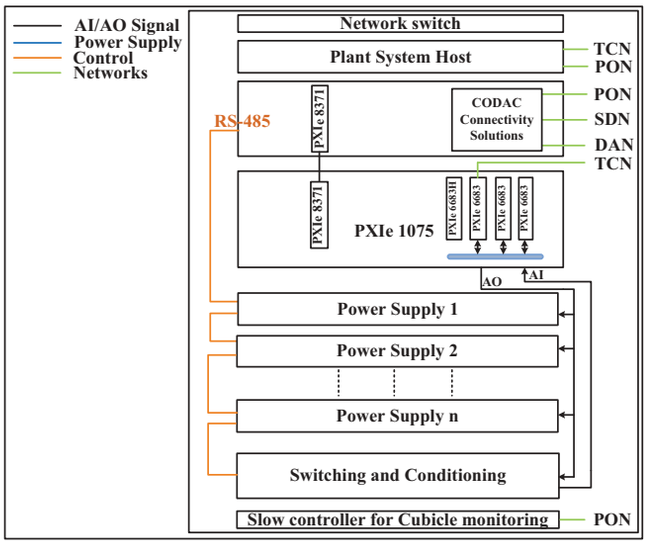


Fig. 7. Architecture of DLP I&C system.

For time synchronization, the fast controller and PSH will receive the absolute time from the TCN. The PSH will set the slow controller timing where the PSH will act as a server of the DLP plant system. All fast controllers will output the control waveform from AO of PXIe 6368 to power supplies at the same time and acquire data with 100 kHz from the MSSCB synchronously, which can satisfy the measurement of the DLP system. Then, the fast controller uploads data to the DAN for fast data archiving and to the SDN for plasma advanced control. Figure 9 is the structure of the DLP fast controller, which includes a HPC, a PXIe 1075 chassis, a PXIe-PCIe8371, a PXIe 6638H time module, and a 3× PXIe 6368.

Since the diagnostic systems provide measurements that are used for advanced machine control, they must be

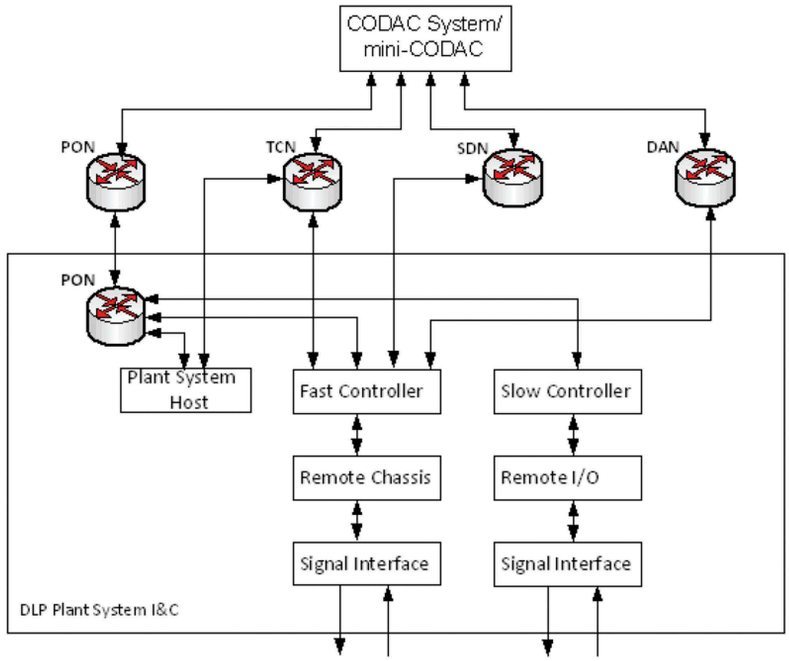


Fig. 8. Network configuration of DLP I&C.

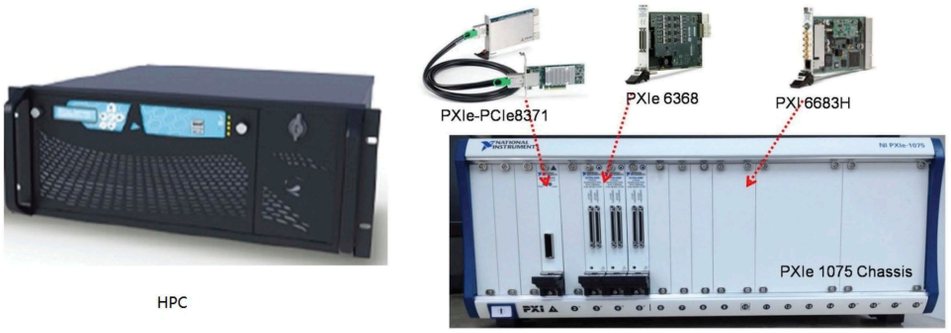


Fig. 9. Structure of DLP fast controller.

operated in a way that guarantees the required measurement accuracy while ensuring high availability. So, an automated operation procedure of the system is designed with the following steps: health checking, configuration, calibration, signal integrity verification, troubleshooting, and start and stop measurements. On the other hand, if a function cannot be automated, it will perform in manual mode by trained operators or plant system experts.

V. CONCLUSIONS

The ITER divertor Langmuir probe system consisting of Langmuir probes, electronics, and I&C is used for machine advanced control and physics study. For the Langmuir probe design, a well-considered and effective structure is used, including only a W probe body, an insulation ceramic, and a complex shape W shield, and the three components are bonded together by special technology developed by XTC for good thermal transfer from the probe to the divertor monoblock. The design of the Langmuir probes is verified by thermal and mechanical analysis under a high heat load in the ITER harsh environment, while the probes also can provide efficient data to satisfy the measurement requirements due to the low probe tip temperature. All back-end electronics are installed in cubicles located at the diagnostic area. The power supply can work at four quadrants and high bandwidth, which can satisfy the recommended DC bias and swept double-probe mode operation. The fast controller sends a control waveform to the amplifier to drive the probes and acquires the analogy signal from the MSSCB with 100 kHz to interpret the I-V curve. The Langmuir probe system design is well considered and shows targeted, robust, and effective characteristics and can meet both physics and engineering requirements.

Acknowledgments

The authors would like to thank the ITER Organization and China Domestic Agency for the support of this work (ITER 5.5.P01.CN.05), providing not only resources but also useful discussion.

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